

Software Requirement Specification (SRS)

Intelligent Water Distribution Monitoring and Leakage Detection Platform

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Assessment: CO2 Assessment Tool 1 – Scenario-Based Requirement Analysis

1. Introduction

The **Intelligent Water Distribution Monitoring and Leakage Detection Platform** is a smart-infrastructure software system designed to help municipal water authorities and utility companies monitor the flow, pressure, and quality of water across a distribution network in real time. The system uses IoT-enabled sensors installed on pipelines, valves, and reservoirs to continuously collect operational data, transmit it to a central platform, and apply analytics to detect leakages, pipe bursts, unauthorized usage, and abnormal consumption patterns at an early stage. The platform aims to reduce non-revenue water (water lost before it reaches the consumer), improve response time to faults, and support data-driven decision-making for maintenance and resource planning.

1.1 Purpose

This document specifies the functional and non-functional requirements of the platform for stakeholders, developers, and testers. It defines the scope, actors, use cases, data requirements, and constraints that will guide the design, development, and validation of the system.

1.2 Business Domain

The system falls under the **Smart Utility / Smart City Infrastructure** domain, specifically water resource management. It integrates concepts from IoT sensing, real-time data analytics, geographic information systems (GIS), and municipal service management.

1.3 Existing Challenges

- Manual leak detection is slow, labour-intensive, and often identifies leaks only after significant water loss or visible surface damage has occurred.
- Utilities lack real-time visibility into pipeline pressure and flow, making it difficult to detect bursts or illegal tapping quickly.
- Significant non-revenue water (NRW) loss due to undetected leakages increases operational cost and impacts water availability for consumers.
- Absence of centralized historical data hampers predictive maintenance and infrastructure planning.
- Manual complaint-based reporting by citizens leads to delayed repairs and poor service quality.

1.4 System Objectives

- Provide continuous, real-time monitoring of flow rate, pressure, and water quality across the distribution network.

- Automatically detect and localize leakages or pipe bursts using sensor data analytics.
- Send instant alerts and notifications to maintenance teams and administrators.
- Enable citizens to report visible leaks and view service updates through a mobile/web application.
- Provide dashboards and reports to support planning, maintenance scheduling, and resource optimization.

2. Problem Description

Municipal water boards manage extensive underground pipeline networks spanning entire cities or regions. Leakages, whether due to ageing infrastructure, pressure fluctuations, corrosion, or illegal tapping, often go unnoticed for extended periods because there is no automated mechanism to detect abnormal flow or pressure drops. This leads to substantial water wastage, increased operational costs, infrastructure damage, and reduced water supply reliability for end consumers. There is a need for an intelligent platform that continuously senses pipeline conditions, applies analytics to distinguish normal usage variation from genuine leaks, and notifies the right personnel with the precise probable location of the fault so that repairs can be carried out swiftly.

3. Scope of the System

The platform will cover sensor data acquisition, real-time transmission, leak-detection analytics, alerting, a web-based administrative dashboard, and a citizen-facing mobile application. It will interface with IoT sensor hardware (flow meters, pressure sensors, acoustic/vibration sensors, water-quality sensors) deployed across the pipeline network and with GIS mapping services for spatial visualization of the network and fault locations.

In Scope:

- Real-time sensor data collection and transmission via IoT gateways.
- Leak and burst detection using flow/pressure anomaly analytics.
- GIS-based visualization of the pipeline network and fault locations.
- Alert and notification management for maintenance teams.
- Citizen complaint reporting and status tracking (mobile/web app).
- Historical data storage, reporting, and analytics dashboards.
- User and role management (admin, field technician, citizen, supervisor).

Out of Scope:

- Physical installation and maintenance of sensor hardware.
- Automated physical repair or robotic pipe-repair mechanisms.
- Billing and revenue collection systems (may integrate via external interface only).

4. Stakeholders

- **Water Utility Administrator** – oversees the overall system, network configuration, and reporting.
- **Field Maintenance Technician** – receives alerts, inspects, and repairs leaks/bursts.
- **Supervisor / Operations Manager** – monitors performance, allocates work orders, reviews analytics.
- **Citizens / Consumers** – report visible leaks and track complaint resolution.
- **IoT Sensor Network** – automated data source (system actor).
- **Municipal Authority / Government Body** – consumes summary reports for policy and budget decisions.

5. Functional Requirements

ID	Requirement	Description
FR-1	Sensor Data Acquisition	The system shall continuously receive flow, pressure, and quality data from IoT sensors at configurable intervals.
FR-2	Real-Time Monitoring Dashboard	The system shall display live pipeline status, sensor readings, and network maps to administrators and supervisors.
FR-3	Leak/Burst Detection	The system shall analyze sensor data to detect abnormal flow or pressure patterns indicating leaks or bursts.
FR-4	Fault Localization	The system shall estimate and display the probable geographic location of a detected leak on the GIS map.
FR-5	Alert & Notification	The system shall send SMS/email/push notifications to relevant technicians and supervisors upon fault detection.
FR-6	Work Order Management	The system shall allow supervisors to create, assign, and track repair work orders.
FR-7	Citizen Complaint Reporting	The system shall allow citizens to report visible leaks with location and photo via a mobile/web app.
FR-8	Complaint Status Tracking	The system shall allow citizens to track the status of their submitted complaints.
FR-9	Historical Data & Reporting	The system shall store historical sensor and event data and generate analytical reports (daily/weekly/monthly).
FR-10	User & Role Management	The system shall support role-based access control for administrators, technicians, supervisors, and citizens.
FR-11	Water Quality Monitoring	The system shall monitor water quality parameters (pH, turbidity, chlorine levels) and flag unsafe readings.
FR-12	System Configuration	The system shall allow administrators to configure sensor thresholds, zones, and network topology.

6. Non-Functional Requirements

Category	Requirement
Performance	The system shall process incoming sensor data and raise leak alerts within 30 seconds of anomaly detection.
Scalability	The system shall support scaling from thousands to hundreds of thousands of sensor nodes without performance degradation.
Reliability	The system shall maintain 99.5% uptime and ensure no data loss during network interruptions using buffered/queued transmission.
Security	The system shall encrypt sensor data in transit and at rest, and enforce role-based authentication and authorization.
Usability	The dashboard and mobile app shall provide an intuitive interface usable by non-technical field staff and citizens.

Category	Requirement
Maintainability	The system architecture shall be modular to allow independent updates to analytics, sensor integration, and UI components.
Availability	Critical monitoring services shall be available 24/7 with redundant server infrastructure.
Interoperability	The system shall support standard IoT communication protocols (MQTT, CoAP, HTTP/REST) for sensor integration.
Data Accuracy	Leak-detection analytics shall maintain a false-positive rate below an agreed threshold (e.g., under 10%).

7. Actors and Use Cases

7.1 Primary Actors

- **Administrator** – configures the system, manages users and zones.
- **Field Technician** – receives alerts, updates repair status.
- **Supervisor** – reviews analytics, assigns work orders.
- **Citizen** – reports leaks, tracks complaints.

7.2 Secondary Actors

- **IoT Sensor Network** – supplies real-time flow, pressure, and quality data.
- **SMS/Email Gateway** – delivers notifications.
- **GIS Mapping Service** – provides spatial mapping for fault localization.

7.3 Use Case Summary

Use Case	Primary Actor	Description
Monitor Network Status	Administrator/Supervisor or	View real-time flow, pressure, and quality data on the dashboard.
Detect Leak/Burst	System (IoT Sensor Network)	Analyze sensor data streams to identify anomalies indicating a leak.
Generate Alert	System	Notify technicians/supervisors when a leak or unsafe reading is detected.
Assign Work Order	Supervisor	Create and assign a repair task to a field technician.
Update Repair Status	Field Technician	Mark a work order as in-progress or completed.
Report Leak	Citizen	Submit a leak report with location and optional photo.
Track Complaint	Citizen	View the current status of a submitted complaint.
View Reports	Administrator/Municipal Authority	Generate and view historical/analytical reports.
Configure Sensors/Zones	Administrator	Set thresholds, zones, and network topology.

7.4 Use Case Diagram (Description)

A UML Use Case Diagram should be drawn with the system boundary labelled "Intelligent Water Distribution Monitoring and Leakage Detection Platform." Place **Administrator**, **Supervisor**, **Field Technician**, and **Citizen** as primary actors on the left, and **IoT Sensor Network**, **SMS/Email Gateway**, and **GIS Mapping Service** as secondary actors on the right. Connect each actor to their respective use cases listed in the table above, using <<include>> relationships from "Detect Leak/Burst" to "Generate Alert," and <<extend>> from "Report Leak" to "Track Complaint."

8. Data Requirements

- **Sensor Data:** timestamp, sensor ID, location (lat/long), flow rate, pressure, water-quality readings.
- **Network Topology Data:** pipeline segments, valves, reservoirs, and zone boundaries (GIS layers).
- **Alert/Event Data:** event ID, type (leak/burst/quality), detected location, severity, timestamp, status.
- **Work Order Data:** work order ID, assigned technician, status, resolution notes, timestamps.
- **Complaint Data:** complaint ID, citizen details, location, description, photo, status history.
- **User Data:** user ID, role, contact details, authentication credentials.
- **Historical/Analytical Data:** aggregated consumption trends, leak frequency by zone, response-time metrics.

9. External Interface Requirements

9.1 User Interfaces

- Web-based administrator/supervisor dashboard with maps, charts, and alert panels.
- Mobile application for field technicians (work order updates) and citizens (leak reporting, complaint tracking).

9.2 Hardware Interfaces

- IoT gateways connecting flow meters, pressure sensors, and water-quality sensors to the cloud platform.

9.3 Software Interfaces

- MQTT/REST APIs for sensor data ingestion.
- GIS mapping service API (e.g., for spatial visualization).
- SMS/Email gateway API for notifications.
- Optional integration with the utility's existing billing/ERP system via REST API.

9.4 Communication Interfaces

- Secure HTTPS/TLS channels for all data transmission between sensors, mobile apps, and the central server.

10. System Features

- **Real-Time Dashboard:** Live visualization of network health, sensor status, and active alerts on a GIS map.
- **AI-Based Anomaly Detection:** Machine-learning models trained on historical flow/pressure data to distinguish genuine leaks from normal demand fluctuation.
- **Automated Alerting:** Multi-channel (SMS/email/push) alerts with severity classification.
- **Citizen Engagement Portal:** Mobile-friendly leak-reporting and complaint-tracking interface.
- **Analytics & Reporting Module:** Zone-wise leak frequency, water-loss estimation, and maintenance performance reports.
- **Role-Based Access Control:** Distinct permission levels for administrators, supervisors, technicians, and citizens.

11. Assumptions and Constraints

11.1 Assumptions

- IoT sensors are installed and functional at critical points across the pipeline network prior to system deployment.
- A stable internet/cellular network is available for sensor-to-cloud data transmission.
- Field technicians and citizens have access to smartphones capable of running the mobile application.
- The utility has authority to access GIS mapping data for the service area.
- Historical consumption data is available in a usable format for training analytics models.

11.2 Constraints

- **Technical:** Sensor hardware has limited battery life and bandwidth, constraining the frequency of data transmission.
- **Operational:** The system depends on timely technician response; delayed field action can affect measured leak-resolution performance.
- **Economic:** Budget limits the number of sensors that can be deployed initially, requiring phased rollout by priority zones.
- **Legal:** The system must comply with data-protection regulations for citizen personal information collected through complaint reporting.
- **Time:** The project must be delivered within the academic/organizational timeline allocated for this assessment/phase.

12. Requirement Validation and Consistency Check

The requirements listed above were cross-checked against the stakeholders and objectives identified in Section 1 to confirm coverage.

12.1 Stakeholder Coverage

- Administrator needs (configuration, reporting) – covered by FR-10, FR-12, and the Reporting/System Features sections.
- Field Technician needs (alerts, work orders) – covered by FR-5, FR-6, and the Update Repair Status use case.
- Supervisor needs (oversight, assignment) – covered by FR-6, FR-9, and Monitor Network Status use case.
- Citizen needs (reporting, tracking) – covered by FR-7, FR-8, and the Citizen Engagement Portal feature.

12.2 Ambiguity, Redundancy, and Inconsistency Check

- No conflicting requirements were identified between real-time monitoring (FR-2) and historical reporting (FR-9), as they operate on separate data views (live vs. aggregated).
- The term "leak" was standardized to distinguish minor seepage, major bursts, and quality violations to avoid ambiguity in alert severity classification.
- Overlap between FR-3 (Leak/Burst Detection) and FR-11 (Water Quality Monitoring) was resolved by treating them as separate analytic pipelines feeding a common alerting module (FR-5).

12.3 Suggested Improvements

- Add a requirement for predictive maintenance scheduling based on pipe-age and failure-history analytics.
- Introduce an offline data-buffering requirement for sensors in low-connectivity zones to prevent data loss.

- Define explicit non-functional thresholds (e.g., exact false-positive rate, exact response-time SLA) in consultation with the utility to remove remaining ambiguity.

End of Software Requirement Specification Document.